

Answer B

$$34) \quad \frac{V_s}{V_p} = \frac{N_s}{N_p} \text{ and } I_s V_s = I_p V_p$$

$$\text{first transformer output } V_{s1} = \frac{N_s}{10} \times V_p ; \text{ second transformer output } V_{s2} = \frac{N_s}{30} \times V_p$$

$$V_{s2} : V_{s1} = 1 : 3$$

$$\text{Hence, } I_{s2} : I_{s1} = 3:1$$

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$$(6.67 \times 10^{-34} \times 2.0 \times 10^8)$$